

COURSE TITLE : PROCESS CONTROL
COURSE CODE : 5063
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER : 52
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Process control in mixing & spinning preparatory	13
2	Process Control in spinning	13
3	Process Control in weaving preparation	13
4	Process control in weaving, knitting & machine balancing	13
TOTAL		52

COURSE OUTCOME

Module.	GO	Student will be able to
1	1	Understand the various norms in process control & yarn realization
	2	Know the Process control in mixing & spinning preparatory
2	1	Understand the productivity and snap reading technique in spinning
	2	Know the Process control in Spinning department
3	1	Understand the Process control in cone winding & warping department
	2	Know the Process control in sizing and Pirn winding
4	1	Understand the Process control in weaving department
	2	Know the Process control in knitting
	3	Understand the Balancing machinery in spinning and weaving

SPECIFIC OUTCOME

MODULE I PROCESS CONTROL IN MIXING & SPINNING PREPARATORY

1.1.0 Understand the various norms in process control & yarn realization

- 1.1.1 State the objectives of process control
- 1.1.2 Define the term “norms”
- 1.1.3 List the methods of establishing norms
- 1.1.4 Define the term quality in terms of yarn and fabric
- 1.1.5 Describe the factors responsible for increasing productivity
- 1.1.6 Identify suitable measures to minimize cost of productivity

- 1.1.7 List the duties and responsibilities of spinning and weaving managers
- 1.1.8 State the importance of machinery auditing
- 1.1.9 Describe the method of controlling yarn realization
- 1.1.10 State the method of maintaining the record for yarn realization and waste
- 1.1.11 Specify the norms for yarn realization
- 1.1.12 Derive the yarn realization in a textile mill

1.2.0 Know the Process control in Mixing & Spinning preparatory

- 1.2.1 State the factors influencing mixing quality and cost
- 1.2.2 Evaluate important physical properties of cotton fibbers
- 1.2.3 Control the mixing quality through fiber characteristics
- 1.2.4 Explain the ATIRA'S approach for linear programming for cotton mixing
- 1.2.5 Describe the method of controlling waste and cleaning in blow room and cards
- 1.2.6 Determine the trash content and cleaning in blow room and cards
- 1.2.7 Analyze the performance of a blow room
- 1.2.8 Identify the machines with substandard performance
- 1.2.9 Explain the method of controlling comber waste
- 1.2.10 Describe the procedure for the control of comber waste
- 1.2.11 Explain the control of sliver evenness
- 1.2.12 State the control of stretch at fly frames
- 1.2.13 Identify suitable measures to control the variability in blow room and Draw frame

MODULE II PROCESS CONTROL IN SPINNING

2.1.0 Understand the productivity and snap reading technique in spinning

- 2.1.1 Define the term productivity
- 2.1.2 Relate productivity with profitability
- 2.1.3 State the the methods for maximizing efficiency in ring spinning
- 2.1.4 Identify the reason for end breaks in spinning and suggest methods for controlling it
- 2.1.5 Discuss the method of measuring productivity
- 2.1.6 Define snap reading
- 2.1.7 Explain the technique of snap reading

2.2.0 Know the Process control in Spinning department

- 2.2.1 Define the term within & between bobbin yarn count variations
- 2.2.2 State the methods of reducing within and between bobbin count variations
- 2.2.3 Explain the control of Count of yarn, Evenness & Variability in lea strength
- 2.2.4 List the different types of yarn irregularities
- 2.2.5 Describe suitable methods to reduce yarn imperfections
- 2.2.6 Indicate the reasons for thick and thin places

MODULE III PROCESS CONTROL IN WEAVING PREPARATION

3.1.0 Understand the Process control in cone winding & warping department

- 3.1.1 Determine the quality of knot
- 3.1.2 Discuss the method of producing good packages in weaving Preparatory
- 3.1.4 Describe the possible defects in cones

- 3.1.5 Define term clearing efficiency and knot factor
- 3.1.6 Discuss the quality factor of yarn clearer
- 3.1.7 Indicate the effects of unwinding tension
- 3.1.8 Suggest suitable measures to minimize end breaks in warping
- 3.1.9 Explain the method of producing quality warp
- 3.1.10 Explain the conditions which influence quality of warper's beam
- 3.1.11 Describe the importance of length measuring motion
- 3.1.12 Explain the method by which the productivity can be improved at Warp winding & warping

3.2.0 Know the Process control in sizing and pirn winding

- 3.2.1 List the objectives of process control in sizing
- 3.2.2 Explain the selection of size and size pick – up
- 3.2.3 State the control of size pick –up
- 3.2.4 Indicate the method of controlling Yarn stretch & moisture in sized yarn
- 3.2.5 Identify the possible defects in sizing
- 3.1.6 Describe the reasons for stoppages in pirn winding
- 3.1.7 Describe the method by which the productivity can be improved at sizing & pirn winding

MODULE IV PROCESS CONTROL IN WEAVING ,KNITTING & MACHINE BALANCING

4.1.0 Understand the Process control in weaving department

- 4.1.1 Select the heald and reed for a particular sort of fabric
- 4.1.2 Discuss the effect of reed parameters on weavability of yarn
- 4.1.3 Suggest suitable measures to improve productivity in loom shed
- 4.1.4 Indicate steps for reducing loom stoppages
- 4.1.5 List the miscellaneous loom stops
- 4.1.6 Suggest measures to control loss of efficiency based on snap reading
- 4.1.7 Indicate procedure for loom allocation
- 4.1.8 Explain the method by which the waste can be minimized in Weaving shed
- 4.1.9 Describe the method of selection of accessories for weaving department

4.2.0 Know the Process control in knitting

- 4.2.1 List different types of needles in knitting
- 4.2.2 Suggest suitable measures to improve productivity in knitting shed
- 4.2.3 Explain the method by which the waste can be minimized in knitting shed
- 4.2.4 Discuss various parameters in warp & weft knitting

4.3.0 Understand the Balancing machinery in spinning and weaving

- 4.3.1 Select the correct number of machines for manufacturing a given quantity of material in spinning and weaving departments
- 4.3.2 Identify the various factors to be considered in estimating the cost of yarn
- 4.3.3 Discuss with specific examples the method of estimating the cost of yarn & cloths
- 4.3.4 Estimate the quantity of warp and weft in a cloth
- 4.3.5 Estimate the quantity of warp and weft required to produce a cloth of given specifications

CONTENT DETAILS

MODULE I PROCESS CONTROL IN MIXING & SPINNING PREPARATORY

Object of process control- “norm , methods of establishing norms - quality - Productivity , factors responsible for increasing productivity , measures to minimize cost of productivity- duties and responsibilities of spinning and weaving managers - machinery auditing - yarn realization - method of maintaining the record for yarn realization and waste - norms for yarn realization - factors influencing mixing quality and cost - Control the mixing quality through fiber characteristics - ATIRA’S approach for linear programming for cotton mixing - method of controlling waste and cleaning in blow room and cards - trash content and cleaning in blow room and cards - performance of a blow room - machines with substandard performance - control of comber waste - control of sliver evenness - control of stretch at fly frames - control of the variability in blow room and Draw frame

MODULE II PROCESS CONTROL IN SPINNING

Productivity - productivity and profitability - maximizing efficiency in ring spinning - end breaks in spinning and control measures - measurement of productivity - snap reading - techniques of snap reading - Within bobbin yarn count variations & Between bobbin yarn count variations, controlling measures - control of Count of yarn, Evenness & Variability in lea strength - yarn irregularities - yarn imperfections ,control measures - thick and thin places, causes and control measures

MODULE III PROCESS CONTROL IN WEAVING PREPARATION

Quality of knot – production of good packages in weaving Preparatory - defects in cones - clearing efficiency and knot factor - quality factor of yarn clearer - unwinding tension - measures to minimize end breaks in warping- production quality warp - conditions which influence quality of warper’s beam - length measuring motion - methods to improve productivity at Warp winding & Warping - objectives of process control in sizing - selection of size and size pick – up - control of size pick –up – control of Yarn stretch & Moisture in sized yarn - defects in sizing - reasons for stoppages of pirn winding - methods to improve productivity Sizing & Pirn Winding

MODULE IV PROCESS CONTROL IN WEAVING , KNITTING & MACHINE BALANCING

Selection of heald and reed for a particular sort of fabric - types of healds and reeds - effect of reed parameters on weavability of yarn - measures to improve productivity in loom shed - steps for reducing loom stoppages - miscellaneous loom stoppages – control of loss of efficiency based on snap reading - procedure for loom allocation – methods to minimize waste in weaving shed - accessories for weaving department and method of selection - types of needles in knitting - measures to improve productivity in knitting shed - methods to minimize waste in knitting shed - parameters in warp & weft knitting - selection of machines for manufacturing a given quantity of material in spinning and weaving departments - various factors to be considered in estimating the cost of yarn - method of estimating the cost of Yarn & Cloth with example – Estimation of the quantity of warp and weft in a cloth – Estimation of the quantity of warp and weft required to produce a cloth of given dimensions

REFERENCE BOOKS

1. A.R.Garg & T.A. Subramanian ATIRA Process control in spinning
2. M.C.Paliwal & P.D.Kimthi ATIRA Process control in weaving
3. SITRA Quality Control in Spinning
4. OMEROD Modern preparation and weaving machinery– BUTTERWORTH